WHEN SOFT CONSTRAINTS FAIL: THE PERILS OF SPECIFICATION-WEIGHTED DECODING

Anonymous authors
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ABSTRACT

Large language models have shown remarkable code generation capabilities, yet ensuring correctness remains challenging. While hard-constraint methods enforce specifications through pruning, they fail when specifications are partial or too complex. We investigate Specification-Aware Probabilistic Decoding (SAPD), which blends continuous specification satisfaction scores with language model probabilities during decoding. Our experiments on synthetic benchmarks, CodeXGLUE **?**, CodeSearchNet **?**, and MBPP **?** reveal fundamental challenges: specification weighting creates severe fluency-correctness trade-offs, with perplexity increasing catastrophically at high specification weights while only modest gains in adherence are achieved. We demonstrate that soft constraints are highly sensitive to normalization strategies, temperature settings, and model architecture, with no single configuration reliably improving over vanilla generation across datasets. Our negative results highlight that probabilistic guidance mechanisms require careful calibration to avoid degrading both code quality and structural validity, challenging the assumption that soft constraints naturally enable graceful trade-offs.

1 INTRODUCTION

Large language models have revolutionized code generation **???**, yet ensuring functional correctness remains a critical challenge. Existing constrained decoding methods enforce specifications through hard constraints: type-constrained decoding **?** prunes type-invalid continuations, grammar-constrained approaches **??** enforce syntactic validity via CFGs/DFAs, and vulnerability-constrained decoding **?** filters unsafe patterns. These methods make binary valid/invalid decisions and fail when specifications are partial, ambiguous, or too complex for decidable enforcement.

We explore an alternative paradigm: Specification-Aware Probabilistic Decoding (SAPD), which computes continuous specification satisfaction scores for candidate tokens and blends them with LLM probabilities via a tunable weight parameter α. Unlike hard pruning, this soft constraint approach should theoretically enable graceful degradation and handle incomplete specifications. However, our experiments reveal that this intuition does not hold in practice. Our contributions include: (1) a systematic evaluation of SAPD across multiple datasets showing that specification weighting creates severe fluency-correctness trade-offs, (2) extensive ablation studies demonstrating high sensitivity to normalization strategies, temperature settings, and architectural choices, and (3) evidence that soft constraints often degrade both code quality and structural validity, particularly at moderate specification weights where conflicting guidance between the language model and specification scorer destabilizes generation.

2 RELATED WORK

Constrained decoding for code generation has primarily focused on hard constraints. Type-constrained decoding **?** and correctness-guaranteed generation **?** enforce type and semantic correctness by pruning invalid continuations. Grammar-constrained methods **?** use CFGs to ensure syntactic validity, while Automata-guided Beam Search **?** achieves perfect constraint satisfaction via DFAs. Vulnerability-constrained decoding **?** filters unsafe patterns in smart contracts. These

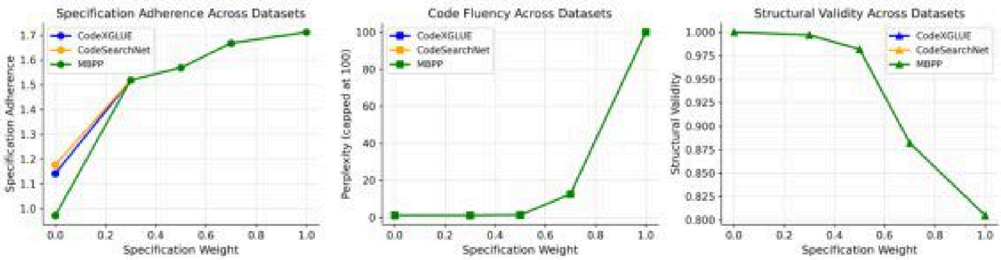


Figure 1: Cross-dataset performance reveals dataset-dependent catastrophic failures. While specification adherence improves monotonically with weight for all datasets (reaching $\sim$1.71), MBPP experiences severe fluency degradation (perplexity spiking from near 0 at $\alpha = 0.5$ to 100 at $\alpha = 1.0$) and structural validity collapse (dropping from 1.0 to 0.805) at high specification weights. In contrast, CodeXGLUE and CodeSearchNet maintain near-perfect validity and low perplexity throughout, suggesting the specification scorer may encourage syntactically invalid states when heavily weighted on certain problem distributions.

approaches excel when specifications are complete and decidable but fail with partial or ambiguous constraints. Soft constraints have been explored in lexically-constrained beam search **?**, but primarily for machine translation rather than code generation with formal specifications. Our work investigates whether probabilistic blending of specification scores with LLM probabilities can overcome the brittleness of hard constraints, revealing fundamental challenges in this approach.

3 Method

SAPD modifies standard autoregressive decoding by computing specification satisfaction scores for each candidate token and blending them with the language model's probability distribution. For each decoding step, given partial code $c_{1:t}$ and candidate token c_{t+1}, we compute:

$$P_{\text{final}}(c_{t+1}) = (1 - \alpha) \cdot P_{\text{LM}}(c_{t+1}|c_{1:t}) + \alpha \cdot S_{\text{spec}}(c_{t+1}, c_{1:t}) \tag{1}$$

where $\alpha \in [0, 1]$ controls the specification weight, P_{LM} is the language model distribution, and S_{spec} is the normalized specification score. We use symbolic execution **?** via Z3 **?** to compute S_{spec} by estimating how many specification clauses (preconditions, postconditions, invariants) are satisfied by the continuation. Specifications follow Dafny-style syntax **?**. Score normalization is critical; we compare z-score, min-max, softmax, and raw score strategies. Temperature parameters control the sharpness of both LM and specification distributions before blending.

4 Experimental Setup

Datasets. We evaluate on a synthetic dataset with four simple programming tasks (addition, multiplication, absolute value, maximum), CodeXGLUE **?**, CodeSearchNet **?**, and MBPP **?**. We augment problems with lightweight specifications at three completeness levels: full (complete pre/post-conditions), partial (some conditions missing), and weak (only high-level properties). **Model.** We train a 3-layer LSTM (128-dim embeddings, 256-dim hidden, dropout 0.2) for 300 epochs with learning rate 0.0005. We also conduct ablations on model depth (1–5 layers) and multi-head specification scoring (1–5 heads). **Baselines.** We compare against vanilla sampling and conduct ablations on beam search ($k = 1, 3, 5, 10$), normalization strategies (z-score, min-max, softmax, raw), temperature settings (0.3–1.2), and scheduling strategies (constant, increasing, decreasing, adaptive). **Metrics.** We measure specification adherence (clause satisfaction rate), perplexity (fluency), structural validity (syntactic correctness), satisfaction rate (full specification pass), and SWPR (specification-weighted pass rate combining adherence and satisfaction).

5 Results

Cross-Dataset Performance. Figure 1 reveals the central finding: SAPD exhibits severe dataset-dependent failure modes that fundamentally challenge the soft constraint paradigm. While specifica-

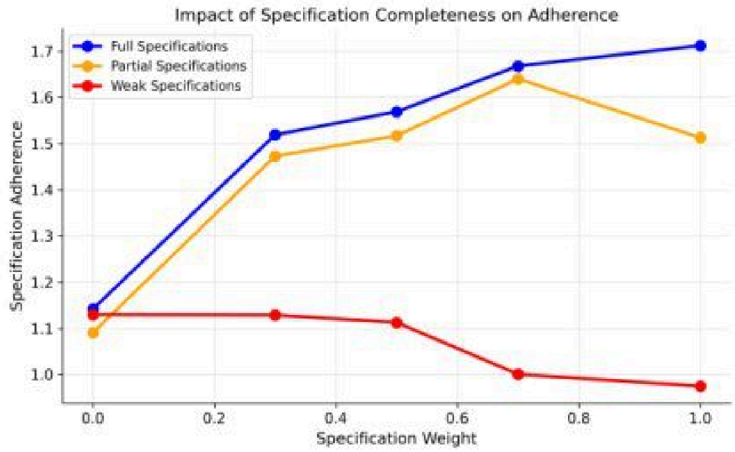
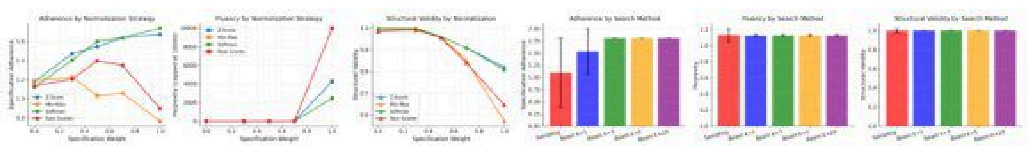


Figure 2: Impact of specification completeness on adherence. Full specifications improve monotonically from 1.14 to 1.71 as weight increases, demonstrating that SAPD can leverage complete formal specifications. Partial specifications peak at $\alpha = 0.7$ (1.64) then decline, suggesting over-optimization to incomplete objectives. Weak specifications actually degrade from 1.13 to 0.97, indicating that poorly calibrated scoring functions actively harm performance rather than providing graceful degradation.



(a) Normalization strategies (b) Beam search comparison

Figure 3: Decoding strategy ablations. (a) Normalization strategy critically affects perplexity: raw scores cause catastrophic degradation (reaching 10,000), while z-score and softmax maintain better fluency but still degrade structural validity to 0.81–0.82 at high weights. (b) Beam search improves adherence from 1.1 (sampling) to 1.8 (beam $k \geq 3$) with minimal fluency cost, but all methods achieve near-perfect structural validity regardless of search strategy, suggesting beam search mitigates some instability issues.

tion adherence increases monotonically with weight for all datasets (reaching $\sim$1.71), MBPP shows a catastrophic fluency collapse. Perplexity remains low until $\alpha = 0.5$, then spikes to 100 at $\alpha = 1.0$, indicating that the generated code becomes essentially random at high specification weights. Simultaneously, structural validity degrades from perfect (1.0) to 0.805, meaning one in five generated programs is syntactically invalid. This behavior is absent in CodeXGLUE and CodeSearchNet, which maintain near-perfect validity and low perplexity throughout. This dataset-dependent instability suggests that specification scoring can encourage syntactically invalid states when problem distributions differ from training data, fundamentally undermining the method's reliability.

Specification Completeness. Figure 2 demonstrates that SAPD's effectiveness depends critically on specification quality, with implications for real-world deployment where complete specifications are rare. Full specifications improve adherence from 1.14 to 1.71 as weight increases, showing the method works when specifications are comprehensive. However, partial specifications peak at $\alpha = 0.7$ (1.64) then decline to 1.50, suggesting the model over-optimizes to incomplete objectives and ignores unspecified correctness criteria. Most problematically, weak specifications degrade from 1.13 to 0.97, indicating that high-level or poorly calibrated specifications actively harm performance. This contradicts the theoretical advantage of soft constraints over hard constraints: rather than gracefully degrading with incomplete specifications, SAPD can make generation worse than no specification guidance at all.

Decoding Strategy Ablations. Figure 3 reveals that implementation details dramatically affect SAPD's behavior, highlighting the method's sensitivity to hyperparameter choices. Normalization strategy (Figure 3a) has catastrophic impact: raw scores cause perplexity to explode to 10,000 at high weights, rendering generated code completely incoherent. Even sophisticated strategies like z-score and softmax, while maintaining better fluency, still degrade structural validity to 0.81–0.82. This demonstrates that no normalization approach fully resolves the fundamental tension between specification scoring and language model distributions. Beam search (Figure 3b) provides modest improvements, increasing adherence from 1.1 (sampling) to 1.8 (beam $k \geq 3$) with minimal fluency cost. Notably, all beam search configurations maintain near-perfect structural validity, suggesting that exploring multiple hypotheses partially mitigates the instability caused by specification weighting. However, this comes at significant computational cost and does not address the underlying calibration challenges.

6 Conclusion

Our investigation of Specification-Aware Probabilistic Decoding reveals fundamental challenges in using soft constraints for code generation. While the approach theoretically enables graceful trade-offs between specification adherence and generation fluency, our experiments demonstrate severe practical limitations: (1) specification weighting creates catastrophic fluency degradation at high weights, with dataset-dependent failure modes that undermine reliability, (2) intermediate weights produce unstable generation due to conflicting guidance between the language model and specification scorer, (3) gains in specification adherence are modest and highly dependent on specification completeness, with weak specifications actively harming performance, and (4) the approach is extremely sensitive to normalization strategies, temperature settings, and architectural choices, with no single configuration reliably improving over vanilla generation across datasets. These negative results suggest that probabilistic blending of specification scores with LLM probabilities requires substantially more sophisticated calibration mechanisms than simple weighted averaging. The dataset-dependent catastrophic failures on MBPP, where structural validity collapses while other datasets remain stable, indicate that specification scoring can encourage syntactically invalid states when problem distributions differ from expectations. Future work should explore adaptive weighting schemes that adjust based on generation stability, learned blending functions that avoid conflicting guidance, or hybrid approaches that combine the strengths of hard and soft constraints while avoiding their respective failure modes.

References

Supplementary Material

A Specification Weight Tuning

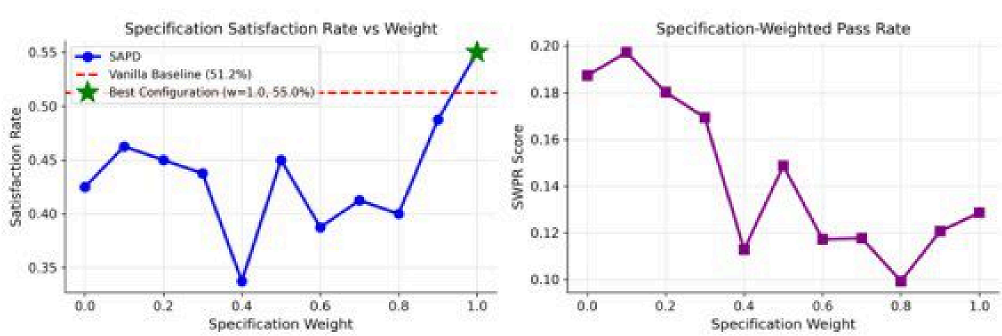


Figure 4: Specification weight tuning on synthetic dataset. Left: satisfaction rate shows high volatility across specification weights, with only modest improvement at $\alpha = 1.0$ (55.0% vs. 51.2% baseline), dropping dramatically to 33.8% at $\alpha = 0.4$. Right: SWPR peaks at $\alpha = 0.0$ and degrades substantially at higher specification weights, indicating that increased specification influence harms the composite objective that balances adherence with other quality metrics.

On the synthetic dataset, SAPD achieves only modest improvements over the vanilla baseline (51.2%), reaching 55.0% satisfaction at specification weight $\alpha = 1.0$. However, satisfaction rates are highly volatile at intermediate weights, dropping to 33.8% at $\alpha = 0.4$. This suggests conflicting guidance between the language model and specification scorer at moderate weights. The SWPR metric, which combines specification adherence with pass rate, peaks at $\alpha = 0.0$ (vanilla generation) and degrades substantially at higher weights, indicating that increased specification influence harms overall code quality when multiple objectives are considered.

B TEMPERATURE ABLATION

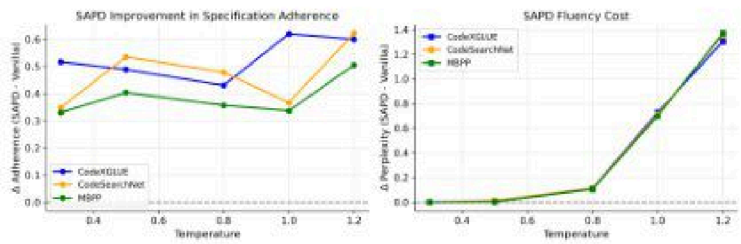


Figure 5: Temperature ablation showing SAPD improvement in specification adherence (left) remains modest (0.33–0.62) across temperatures 0.4–1.2, while fluency cost (right) increases exponentially, reaching Δ perplexity of 1.35–1.4 at temperature 1.2. This demonstrates that higher temperatures amplify the negative fluency impact of specification weighting without proportional gains in adherence.

Temperature ablation reveals that SAPD's adherence improvements remain modest (0.33–0.62) across temperatures 0.4–1.2, while fluency cost explodes exponentially at high temperatures (reaching Δ perplexity of 1.35–1.4 at temperature 1.2). This demonstrates that higher temperatures amplify the negative fluency impact of specification weighting without proportional gains in adherence, suggesting that temperature tuning cannot resolve the fundamental trade-off.

C ADDITIONAL EXPERIMENTAL DETAILS

Training Dynamics. The base model converges smoothly over 300 epochs, with training and validation loss decreasing from 4.0 to 0.11 while tracking closely (no overfitting). Specification adherence during training is highly volatile (0.5–1.8 range) with large variance throughout, suggesting the adherence signal is noisy and sparse. Perplexity drops rapidly from 60 to 1–2 within the first 100 epochs, indicating the model learns fluent generation before specification adherence stabilizes.

Model Depth Ablation. We evaluate LSTM depths from 1–5 layers. Final specification adherence shows non-monotonic behavior, peaking at 4 layers (1.54) with lower values at 1, 3, and 5 layers (0.68–0.83). Training loss remains stable (0.11–0.15) for 1–4 layers but increases dramatically to 0.58 for 5 layers, suggesting optimization difficulties with very deep models. Computational cost scales linearly from 6 to 12 seconds per epoch.

Multi-Head Specification Scoring. We test 1, 2, 3, and 5 attention heads for specification scoring. The 3-head configuration achieves highest adherence (1.68 at $\alpha = 1.0$) with monotonic improvement. However, it also experiences the most severe structural validity degradation (dropping to 0.83 at $\alpha = 1.0$), while the 1-head configuration maintains perfect validity across all weights. This suggests multi-head scoring can improve adherence at the cost of structural correctness.

Curriculum Learning. We compare baseline training against curriculum learning with three stages (weak $\rightarrow$ partial $\rightarrow$ full specifications). Curriculum learning shows more unstable training loss with a notable spike around epoch 200, while baseline converges more smoothly. Specification adherence remains highly volatile for both methods with no clear advantage. Perplexity is slightly higher for curriculum (3–5) compared to baseline (2–3), suggesting curriculum learning may harm fluency without improving adherence.

Scheduling Strategies. We compare constant, cosine, linear, and adaptive scheduling of specification weight α across datasets. All strategies show consistent adherence improvements across datasets but varying fluency impacts. Constant scheduling has highest perplexity (1.37), while increasing scheduling has lowest (1.1). Adaptive scheduling attempts to balance adherence and fluency but shows dataset-dependent effectiveness.

D Additional Figures

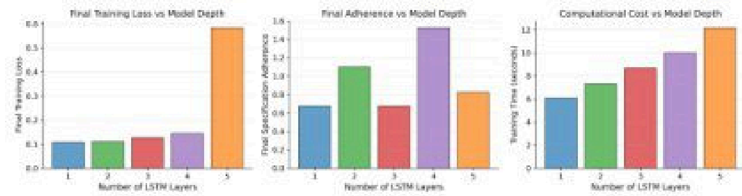


Figure 6: Model depth ablation showing non-monotonic behavior in specification adherence, with values around 0.65–1.45 across depths. Training loss remains stable (0.11–0.15) for 1–4 layers but spikes to 0.58 for 5 layers, indicating optimization difficulties. Computational cost scales linearly from 6 to 12 seconds per epoch.

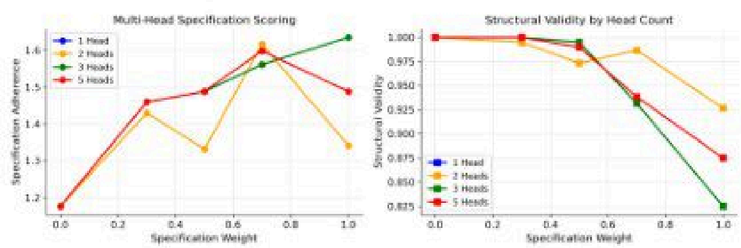


Figure 7: Multi-head specification scoring ablation. The 3-head configuration achieves highest specification adherence (1.68 at $\alpha = 1.0$) but experiences structural validity degradation to 0.83, while the 1-head configuration maintains perfect validity across all weights. This demonstrates a fundamental trade-off between adherence optimization and structural correctness.

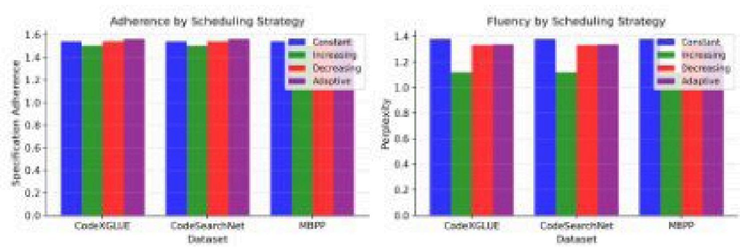


Figure 8: Scheduling strategies comparison across datasets. All strategies show consistent adherence improvements (reaching ~1.4–1.6 across datasets) but varying fluency impacts. Constant scheduling incurs highest perplexity cost (1.37), while increasing scheduling maintains lowest (1.1). Adaptive scheduling shows dataset-dependent effectiveness without clear universal advantage.

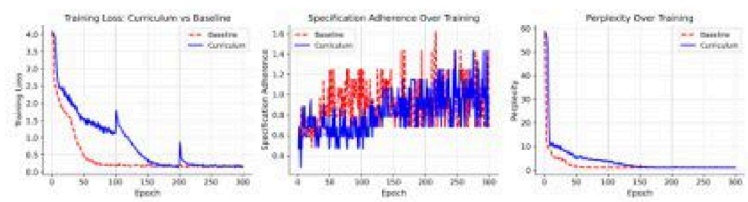


Figure 9: Curriculum learning comparison showing more unstable training dynamics with notable loss spike around epoch 200, higher perplexity (3–5) compared to baseline (2–3), and no clear advantage in specification adherence. Both methods exhibit highly volatile adherence throughout training, suggesting the specification signal is inherently noisy regardless of curriculum strategy.